# Watch Your Teacher Build an AI KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

**1. How long by hand?** 140 cards, 300 passes, 1 minute per card.

140 × 300 = 42,000 cards = 42,000 minutes ≈ 700 hours ≈ 29 days

**2. Six steps to build an AI.** Number the steps in order, 1 to 6.

| | | | |
|---|---|---|---|
| 5 | Train: many passes, fix the weights | 1 | Get data |
| 3 | Split the data: training and test | 6 | Test, then use it |
| 2 | Ask a question | 4 | Build the network |

Why does the network never train on the test data? So we can check it on data it never saw.

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**3. Predict, train, and test.** Write your prediction **before** each run.

| Run | Question | My prediction (test %) | Training % | Test % |
|---|---|---|---|---|
| 1 | Fighting or Psychic? (2 numbers) | Answers vary. | about 90% | about 90% |
| 2 | Fighting or Psychic? (8 numbers) | | near 100% | lower |
| 3 | All 18 types | | about 40% | about 20% |

Scores change a little with each random split. The patterns stay the same.

**Word bank:** data overfitting

### 4. What did you see?

In Run 2, the training score went up, but the test score did not. The network learned the training examples by heart. This is called overfitting.

In Run 3, the network was not much better than guessing. Why? The answer is not in the data.

An AI can only learn what is in its data.

### 5. Talk about it. What if someone used an AI like this to decide who gets a job?

A problem could be Answers vary. It only knows what is in its data; it can be wrong.

### Go Further (Day 2). Ask the network. Try 3 Pokémon you know.

| Pokémon | The network said | Really | Right? | Test Pokémon? |
|---|---|---|---|---|
| Answers vary. | | | | |
| | | | | |
| | | | | |